



BAULKHAM HILLS HIGH SCHOOL

2011
YEAR 11 YEARLY

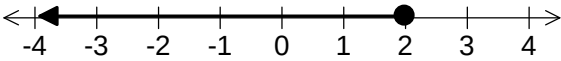
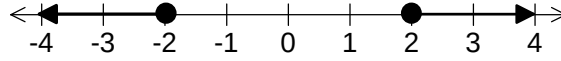
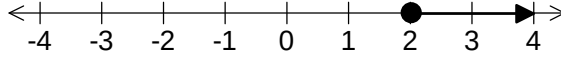
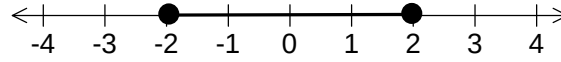
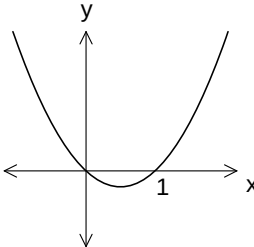
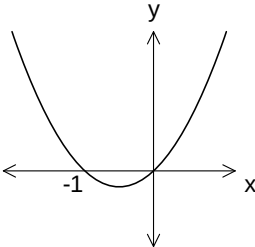
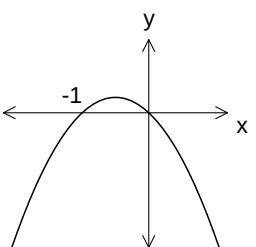
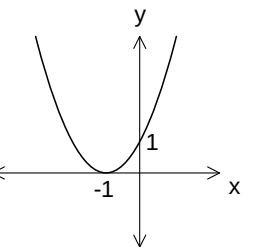
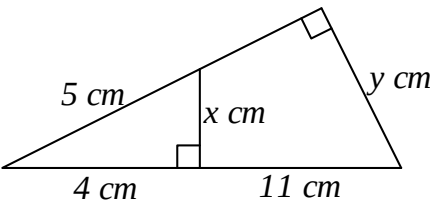
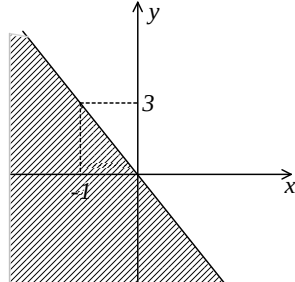
Mathematics Advanced

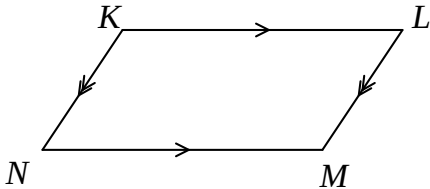
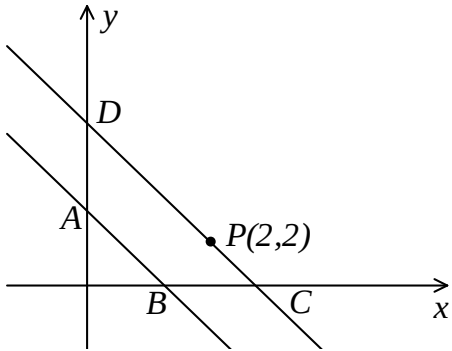
General Instructions

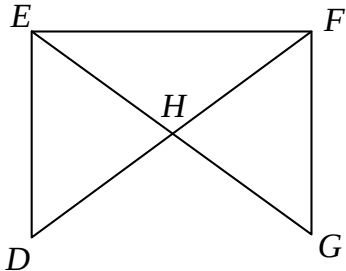
- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work

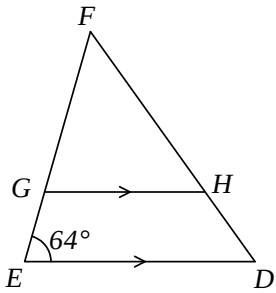
Total marks – 100

- Attempt Questions 1 – 9
- All questions are of equal value
- Start each question on a new sheet of paper.
- Write your name and the question number at the top of each sheet.

Question 1 (10 marks) - Answer the following on the answer sheet provided.	Marks
a) Evaluate $\sqrt{\frac{19.28}{4.25 \times 3.5}}$ correct to 3 significant figures	2
b) Which of the following represents the solution of $x^2 \leq 4$? (A)  (B)  (C)  (D) 	1
c) Which of the following is the graph of $y = x^2 + x$? (A)  (B)  (C)  (D) 	1
d) For which value of x is $ 2x + 5 \leq 7$ <u>false</u> ? (A) -1 (B) 0 (C) 1 (D) 4	1
e) Find the distance AB for $A(-3, -4)$, $B(5, 7)$ to one decimal place	1
Questions (f) and (g) refer to the information below. 	f) The value of x is: (A) 1 (B) 2 (C) 3 (D) 4 g) The value of y is: (A) 4 (B) 6 (C) 8 (D) 9
h) The shaded region is best described by which inequality: (A) $y < -\frac{1}{3}x$ (B) $y \geq -3x$ (C) $y \leq -3x$ (D) $y \leq -\frac{1}{3}x$	1 
i) If a curve is always increasing which <u>must</u> occur for all values of x : (A) $\frac{dy}{dx} > 0$ (B) $\frac{dy}{dx} < 0$ (C) $\frac{d^2y}{dx^2} > 0$ (D) $\frac{d^2y}{dx^2} < 0$	1

Question 2 (11 marks) - Start on a new page		Marks
a)	Factorise fully $x^4 - 27xy^3$	2
b)	Simplify and express your answer as a rational number $\frac{1}{3\sqrt{2} + 1} + \frac{1}{1 - 3\sqrt{2}}$	2
c)	Solve $(y - 5)^2 - 1 = (y - 4)(y + 3)$	2
d)	Find the exact value of $\tan 120^\circ$	2
e)	M is the midpoint of $A(5, k)$ and $B(11, 8)$. Find k , if M lies on the line $x - 2y = 4$	3
Question 3 (13 marks) - Start a new page		Marks
a)	Solve $3^{6x+3} = 27^{x+2}$	2
b)	 In a parallelogram $KLMN$ $\angle NKL = 3x - 30^\circ$ and $\angle KLM = x + 10^\circ$ Find the size of $\angle KNM$ (with working and no reasons required)	3
c)	 The diagram shows the straight line AB with equation $x + 2y = 2$ i) Show that the equation of DC parallel to AB passing through $(2, 2)$ is $x + 2y = 6$ ii) Calculate the perpendicular distance from P to AB iii) Calculate the area of trapezium $ABCD$	2 2 2
d)	Derive the equation of the tangent to the curve $y = 2x^2 + 5$ at the point $(2, 13)$	2

Question 4 (11 marks) - Start a new page		
a)	Solve the inequality $3x^2 - 5x - 12 \geq 0$	2
b)	Differentiate i) $y = \frac{3x+2}{4x-3}$ ii) $f(x) = x\sqrt{x}$ iii) $y = (2x+1)(5x-3)^4$	2 2 2
c)	Show that i) $\cos \theta \tan \theta = \sin \theta$ ii) Hence solve $\cos \theta \tan \theta = -\frac{1}{2} \quad \text{for } 0^\circ \leq \theta \leq 360^\circ$	1 2
Question 5 (11 marks) - Start a new page		Marks
a)	If $y = ax^2 + bx$ has a maximum at (2,3), find the values for a and b	2
b)	 H is the point of intersection of DF and EG. If $DH = GH$ and $\angle HEF = \angle HFE$ Show that $\triangle EHD \equiv \triangle FHG$ (Draw your own diagram and mark all given information)	4
c)	Find the domain and range for $y = \frac{1}{2x-3}$	2
d)	Sketch $y = 9x - 11 - 2x^2$ by first finding the vertex and any intercepts	3

Question 6 (11 marks) - Start a new page		
a)	The roots of the equation $2x^2 - 4x + 1 = 0$ are α and β. Without solving the equation find: i) $\alpha + \beta$ ii) $\alpha\beta$ iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ iv) $\alpha^3 + \beta^3$	1 1 1 2
b)	 Given that $\triangle EDF \parallel \triangle GHF$ i) With reasons, find HF when $ED = 6m$, $GH = 4m$ and $DH = 4m$ ii) Find $\angle HFG$ to the nearest min iii) Find area of $\triangle EDF$ to 1 decimal place	2 2 2

Question 7 (11 marks) - Start a new page		Marks
a)	Find A, B and C such that $3x^2 - x + 3 \equiv A(x - 1)(x + 2) + B(x - 2) + C$	3
b)	The function $f(x)$ is defined by $f(x) = x^3 - 3x^2 - 9x$ i) Find the turning points for the curve $y = f(x)$ and determine their nature ii) Show that there is a point of inflexion and find its coordinates iii) Sketch the graph of $y = f(x)$ showing relevant information for the domain $-3 \leq x \leq 4$ iv) What is the maximum value of this curve in this domain	3 1 3 1

Question 8 (11 marks) - Start a new page		
a)	Find the values of x for which $2x - 5 \leq x + 7 \leq 4x + 1$	2
b)	Solve $4 \sin^2 x = 3$ for $0^\circ \leq x \leq 360^\circ$	2
c)	A garden bed is to be constructed in the shape of a semi-circle added to a rectangle. The width of the rectangle is x and the length y metres as shown in the diagram. Forty metres of edging material is available to use around the edge of the garden bed i) Show that $y = 20 - \frac{1}{2}x - \frac{1}{4}\pi x$ ii) Show that the area of the garden bed can be expressed as $A = 20x - \frac{1}{2}x^2 - \frac{1}{8}\pi x^2$ iii) Find the dimensions of the rectangle such that the area contained in the garden bed is a maximum.	2 1 4

Question 9 (11 marks) - Start a new page		Marks
a)	Show that the constants k and l have different signs, if the quadratic equation: $(k^2 + l^2)x^2 - 2m(k + l)x + m^2 = 0$ has no real roots Where $m \neq 0$	3
b)	$ABCD$ is a square of side x cm, with P a point within the square such that $PC = 6\text{cm}$, $PB = 2\text{cm}$ and $AP = 2\sqrt{5}\text{cm}$. Let $\angle PBC = \alpha$ i) By using triangle PBC show that $\cos \alpha = \frac{x^2 - 32}{4x}$ ii) By considering triangle PBA, show that $\sin \alpha = \frac{x^2 - 16}{4x}$ iii) Hence or otherwise, show that the value of x is a solution of $x^4 - 56x^2 + 640 = 0$ iv) Find x. Give reasons for your answer.	1 2 2 3
End of Exam		

Question one. (10 marks)

a) 1.14 $\begin{cases} \textcircled{2} \text{ for correct answer} \\ \textcircled{1} \text{ for 3 sig fig, or } 1.138... \end{cases}$

b) D c) B d) D e) 13.6 f) C

g) D h) C i) A.

Question two (11 marks)

a) $x^4 - 27xy^3 = x(x^3 - 27y^3)$ — $\textcircled{1}$

$= x(x - 3y)(x^2 + 3y + y^2)$ — $\textcircled{1}$

b) $\frac{1}{3\sqrt{2}+1} + \frac{1}{1-3\sqrt{2}} = \frac{1}{3\sqrt{2}+1} - \frac{1}{3\sqrt{2}-1}$

$= \frac{3\sqrt{2}-1-3\sqrt{2}-1}{9 \times 2 - 1}$ $\textcircled{1}$

$= \frac{-2}{17}$ $\textcircled{1}$

c) $(y-5)^2 - 1 = (y-4)(y+3)$ $\textcircled{1}$

$y^2 - 10y + 24 = y^2 - y - 12$

$-9y = -36$

$y = 4$ $\textcircled{1}$

d) $\tan 120^\circ = -\tan 60^\circ$
 $\textcircled{1} = \frac{\sqrt{3}}{2}$ — $\textcircled{1}$

e) M: $\left(\frac{16}{2}, \frac{k+8}{2}\right) = \left(8, \frac{k+8}{2}\right)$ $\textcircled{1}$

sub in

$8 - 2\left(\frac{k+8}{2}\right) = 4$ $\textcircled{1}$

$8 - k - 8 = 4$

$k = -4$ $\textcircled{1}$

Question 3 (13 marks)

a) $3^{6x+3} = 3^{3x+6}$ ①

$\therefore 6x+3 = 3x+6$

$3x = 3$

$x = 1$ ①

b) $3x - 30 + x + 10 = 180$ ✓ ①

$4x = 200$

$x = 50$ ✓ ①

$\angle KLM = 50 + 10$

$= 60$

$\angle KNM = \angle KLM$

$= 60^\circ$ ✓ ①

c) i) AB: $2y = 2 - x$

$y = 1 - \frac{1}{2}x$

$m_1 = -\frac{1}{2}$ ①

P(2,2)

DC: $y - y_1 = m(x - x_1)$

$y - 2 = -\frac{1}{2}(x - 2)$ ①

$2y - 4 = -x + 2$

$x + 2y = 6$ or $(x + 2y - 6 = 0)$

ii) $pd = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$= \frac{|1 \times 2 + 2 \times 2 - 6|}{\sqrt{1 + 4}}$ ①

$= \frac{4}{\sqrt{5}}$ or $\frac{4\sqrt{5}}{5}$ ①

$x + 2y - 2 = 0$: AB

P(2,2)

$= 1.79$ (2d.p)

iii) Area ABCD

D: (0,3) C: (6,0)

$A_{\Delta_1} = \frac{1}{2} \times 1 \times 2 = 1$ ①

$A_{\Delta_2} = \frac{1}{2} \times 3 \times 6 = 9$

$A_{ABCD} = 9 - 1$

$= 8$ u² ①

or

$\overline{AB} = \sqrt{5}$

$\overline{CD} = \sqrt{9 + 36} = \sqrt{45}$ ①

$A_{trap} = \frac{1}{2} \times \frac{4}{\sqrt{5}} (\sqrt{5} + 3\sqrt{5})$

$= 8$ u² ①

d) $y = 2x^2 + 5$ at (2,7)
 $\frac{dy}{dx} = 4x$ ①
 at $x = 2$ $m = 8$
 $\therefore y - y_1 = m(x - x_1)$ ①
 $y - 7 = 8(x - 2)$ ①
 $y = 8x - 9$

OR if at $x = 2$

$\therefore y = 13$

$\therefore y - 13 = 8(x - 2)$

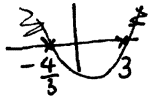
$y = 8x - 3$

Question 4 (11 marks)

a) $(3x+4)(x-3) \geq 0$

$x = -\frac{4}{3}, 3$

$\therefore x \leq -\frac{4}{3}, x \geq 3$



② both answers

① working towards soln.

b) i) $y' = \frac{(4x-3) \times 3 - 4(3x+2)}{(4x-3)^2}$
 $= \frac{12x-9-12x-8}{(4x-3)^2}$

① $= \frac{-17}{(4x-3)^2}$

ii) $f(x) = x\sqrt{x} = x^{\frac{3}{2}}$

$f'(x) = \frac{3}{2}x^{\frac{1}{2}}$ or $\frac{3\sqrt{x}}{2}$

or $f'(x) = \frac{x}{2\sqrt{x}} + \sqrt{x}$

②

iii) $\frac{dy}{dx} = 2(5x-3)^4 + 20(5x-3)^3(2x+1)$
 $= 2(5x-3)^3(5x-3 + 20x + 10)$
 $= 2(5x-3)^3(25x+7)$

c) i) $\cos \theta + \tan \theta = \sin \theta$

L.H.S. = $\cancel{\cos \theta} \times \frac{\sin \theta}{\cancel{\cos \theta}}$ ①

$= \sin \theta = \text{R.H.S.}$

ii) $\cos \theta + \tan \theta = -\frac{1}{2}$

$\therefore \sin \theta = -\frac{1}{2}$

$\theta = 210, 330$
 ① ①

Question 5 (11 marks)

a) $y = ax^2 + bx$

$\rightarrow 3 = 4a + 2b \dots \text{②}$

$\frac{dy}{dx} = 2ax + b = 0$ at stat pt

$(2, 3) : 4a + b = 0 \dots \text{①}$

①

$4a + 2b = 3$ sub ① from ②

$4a + b = 0$

$b = 3$

sub into ①

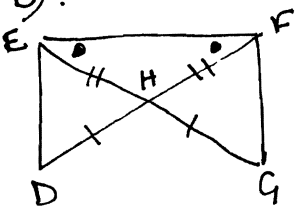
$4a = -3$

$a = -\frac{3}{4}$

$\therefore a = -\frac{3}{4} \quad b = 3$ ①

quest 5 cont.

b).



In $\triangle HEF$

$$\widehat{HEF} = \widehat{HFE} \text{ (given)}$$

$\therefore \triangle$ is isosceles (base $\angle$ s equal) ①

then $EH = HF$ (sides opposite equal $\angle$ s)

In $\triangle EHD$ and $\triangle FHG$

$$EH = FH \text{ (above)}$$

$$\widehat{EHD} = \widehat{FHG} \text{ (vertically opposite } \angle\text{s)}$$

$$DH = HG \text{ (given)}$$

$$\therefore \triangle EHD \cong \triangle FHG \text{ (S.A.S)}$$

③ marks

c) $\mathbb{D}$: all real x , $x \neq \frac{3}{2}$ ①

$\mathbb{R}$: all real y , $y \neq 0$ ①

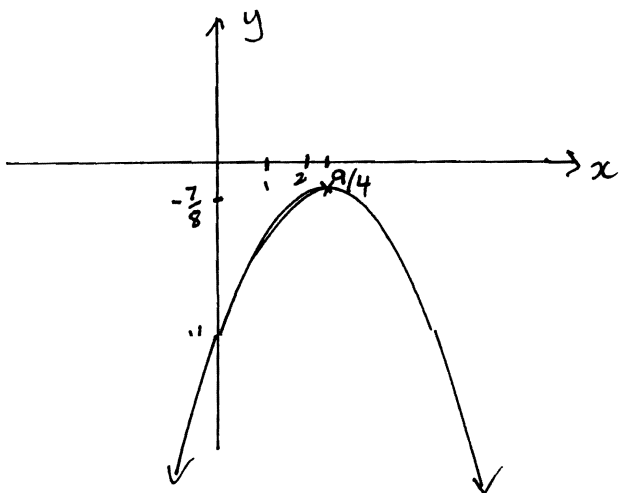
d) $y = 9x - 11 - 2x^2$

axis of symmetry: $x = \frac{-b}{2a}$
 $= \frac{-9}{-4}$
 $= \frac{9}{4}$

$$y = \left(9 \times \frac{9}{4}\right) - 11 - 2 \times \left(\frac{9}{4}\right)^2$$

$$= -0.875$$

Vertex: $\left(\frac{9}{4}, -0.875\right)$
 $\left(\frac{9}{4}, -\frac{7}{8}\right)$



Question 6 (11 marks)

a) $2x^2 - 4x + 1 = 0$

i) $\alpha + \beta = -\frac{b}{a}$
 $= \frac{4}{2} = 2$ ①

ii) $\alpha\beta = \frac{c}{a}$
 $= \frac{1}{2}$ ①

iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta}$
 $= \frac{2}{\frac{1}{2}} = 4$ ①

iv) $\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)$
 $= (\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta)$ ①
 $= 2(2^2 - \frac{3}{2})$
 $= 5$ ①

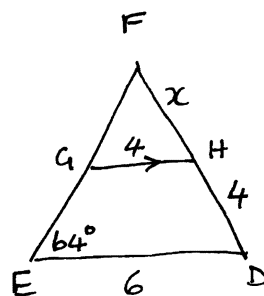
b) i) $\frac{FH}{FD} = \frac{GH}{ED}$ (sides in ratio in similar Δ s) ①

$$\frac{x}{x+4} = \frac{4}{6}$$

$$6x = 4x + 16$$

$$2x = 16$$

$$FH = x = 8 \text{ m}$$
 ①



ii) $\frac{\sin \theta}{4} = \frac{\sin 64}{8}$ ①

$$\sin \theta = \frac{4 \sin 64}{8}$$

$$= 0.44939 \dots$$

$$\theta = 26^\circ 42'$$
 ①

iii) Area $= \frac{1}{2} ab \sin C$
 $= \frac{1}{2} \times 12 \times 6 \times \sin 89^\circ 18'$
 $= 35.997 \dots$
 $= 36.0 \text{ m}^2$ ①

$$\hat{FDE} = 180 - 26^\circ 42' - 64$$

$$= 89^\circ 18'$$
 ①

Question 7 (11 marks)

a) $3x^2 - x + 3 \equiv A(x^2 + x - 2) + B(x - 2) + C$

RHS = $Ax^2 + x(A+B) - 2A - 2B + C$ ①

equating: $3 = A$ $-1 = A+B$ ① $3 = -2A - 2B + C$

$-1 = 3+B$

$3 = -6 + 8 + C$

$B = -4$

$1 = C$

$\therefore A = 3, B = -4, C = 1$ ①

b) $f(x) = x^3 - 3x^2 - 9x$

i) $f'(x) = 3x^2 - 6x - 9$

$3(x^2 - 2x - 3) = 0$

$(x-3)(x+1) = 0$ ①

$x = -1, 3$

$x = -1$

$x = 3$

$y = (-1)^3 - 3(-1)^2 - 9(-1)$

$= 5$

$(-1, 5)$

①

$y = 3^3 - 3 \times 3^2 - 9 \times 3$

$= -27$

$(3, -27)$

$f''(x) = 6x - 6$

test

$f''(-1) = -6 - 6$

$= -12 < 0$

max

$f''(3) = 18 - 6$

$= 12 > 0$

min

①

ii) $f''(x) = 6x - 6 = 0$ for P.O.I

$x = 1$

$y = 1^3 - 3(1) - 9 \times 1$

$= -11$

$(1, -11)$

①

Test

$f''(0) = -6$

$f''(2) = 12 - 6$

$= 6$

$\therefore$ change in concavity

iii) End pts

$-3 = x$

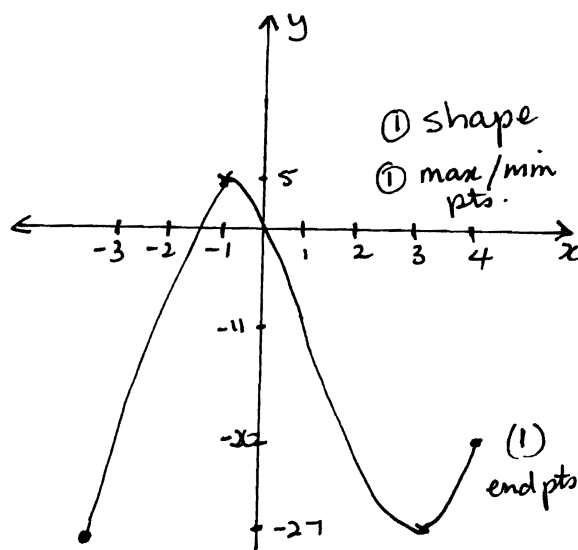
$(-3, -27)$

$x = 4$

$f(4) = -20$

$(4, -20)$

iv) max value is 5 ①



Question 8 11 marks

a) $2x - 5 \leq x + 7 \leq 4x + 1$

$$2x - 5 \leq x + 7$$

$$x \leq 12 \quad \textcircled{1}$$

$$x + 7 \leq 4x + 1$$

$$6 \leq 3x$$

$$2 \leq x$$

$$\therefore 2 \leq x \leq 12 \quad \textcircled{1}$$

b) $4 \sin^2 x = 3$

$$\sin x = \pm \frac{\sqrt{3}}{2} \quad \textcircled{1}$$

$$x = 60, 120, 240, 300 \quad \textcircled{1}$$

c)

i) $P = \frac{1}{2} \pi x + 2y + x \quad \textcircled{1}$

$$40 = 2y + \frac{1}{2} \pi x + x$$

$$2y = 40 - \frac{1}{2} \pi x - x \quad \textcircled{1}$$

$$y = 20 - \frac{1}{2} x - \frac{1}{4} \pi x$$

ii) $A = xy + \frac{1}{2} \pi x \left(\frac{x}{2}\right)^2$

$$= x \left(20 - \frac{1}{2} x - \frac{1}{4} \pi x\right) + \frac{\pi x^3}{8} \quad \textcircled{1}$$

$$= 20x - \frac{1}{2} x^2 - \frac{1}{4} \pi x^2 + \frac{\pi x^3}{8}$$

$$= 20x - \frac{1}{2} x^2 - \frac{\pi x^2}{8}$$

iii) $\frac{dA}{dx} = 20 - x - \frac{\pi x}{4} \quad \textcircled{1}$

$$\text{let } \frac{dA}{dx} = 0$$

$$20 - x - \frac{\pi x}{4} = 0$$

$$20 = x \left(1 + \frac{\pi}{4}\right)$$

$$x = \frac{80}{4 + \pi} \quad \textcircled{1}$$

$$\frac{d^2 A}{dx^2} = -1 - \frac{1}{4} \pi < 0 \quad \textcircled{1}$$

max for all values.

dimensions

$$x = \frac{80}{4 + \pi} = 11.2 \text{ (1dp)}$$

$$y = \frac{40}{4 + \pi} = 5.6 \text{ (1dp)}$$

①

Question 9 (11 marks)

a) $\Delta < 0$ no real roots ①

$$(k^2 + l^2)x^2 - 2m(k+l)x + m^2 = 0$$

$$\Delta = 4m^2(k+l)^2 - 4m^2(k^2 + l^2) < 0$$

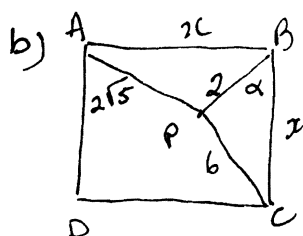
$$(k+l)^2 - (k^2 + l^2) < 0 \quad ①$$

$$k^2 + 2kl + l^2 - k^2 - l^2 < 0$$

$$2kl < 0$$

$$kl < 0$$

$\therefore$ either k or l is negative and $\therefore$ opposite signs



ii) In $\triangle PBC$

$$i) \cos \alpha = \frac{2^2 + x^2 - 6^2}{2 \times 2 \times x} \quad ①$$

$$= \frac{x^2 - 32}{4x}$$

ii) In $\triangle PBA$ $\hat{APB} = 90^\circ - \alpha$

$$\cos(90^\circ - \alpha) = \sin \alpha \quad ①$$

$\therefore$ using cosine rule

$$\sin \alpha = \frac{x^2 + 2^2 - (2\sqrt{5})^2}{2 \times 2 \times x} \quad ①$$

$$= \frac{x^2 + 4 - 20}{4x} = \frac{x^2 - 16}{4x}$$

iii) now $\sin^2 \alpha + \cos^2 \alpha = 1$

$$\left(\frac{x^2 - 16}{4x}\right)^2 + \left(\frac{x^2 - 32}{4x}\right)^2 = 1 \quad ①$$

$$(x^2 - 16)^2 + (x^2 - 32)^2 = 16x^2$$

$$x^4 - 32x^2 + 256 + x^4 - 64x^2 + 1024 = 16x^2$$

$$2x^4 - 112x^2 + 1280 = 0 \quad ①$$

$$\therefore x^4 - 56x^2 + 640 = 0$$

iv) $x^2 = \frac{56 \pm \sqrt{56^2 - 4 \times 640}}{2}$

$$= \frac{56 \pm 24}{2} \quad ①$$

$$= 40, 16$$

$$\therefore x = \pm \sqrt{40} \pm 4$$

but $x > 0$ — for length

$$x = \sqrt{40}, 4 \quad ①$$

For α to be acute $x^2 > 32$ — or $\cos \alpha = \frac{x^2 - 32}{4x}$ would be obtuse ①

$$x = \sqrt{40} = 2\sqrt{10}$$

alternate
(if $x = 4$ $\sin \alpha = 0, \alpha = 0$ } no solution here.
 $\cos \alpha = -1, \alpha = 180^\circ$

if $x = \sqrt{40}$ $\sin \alpha = \frac{3}{\sqrt{10}}, \alpha = 71^\circ 34'$ OK.
 $\cos \alpha = \frac{1}{\sqrt{10}}, \alpha = 71^\circ 34'$